

Cashfree Checkout - Web

We need API Keys for this. You can get that from:

<https://merchant.cashfree.com/merchants/pg/developers/api-keys?env=test>

Now we need to follow the documentation:

<https://docs.cashfree.com/docs/webintegration-best-practices>

<https://www.cashfree.com/devstudio>

We are going to select Web where you can choose UI options.

For this documentation, I am going with **Redirect Checkout**.

<https://www.cashfree.com/devstudio/preview/pg/web/checkout>

1. Setup

• Config

Select the environment and fill the API Keys (App ID and Secret Key) to get started. [Read Docs](#)

[Click here](#) to try with sample App ID and Secret Key

Environment	App ID	Secret Key
Sandbox	TEST430329ae80e0f32e41a393d78b92	TESTaf195616268bd6202eeb3bf8dc458956e7192i
API Version	API URL	
2023-08-01 (Latest)	https://sandbox.cashfree.com/pg	

Next

Here as well you can get the API keys by clicking on the green text “Click here”

Make sure your options are filled just like this and api keys will be different.

2. Initialize SDK

Initialize Cashfree JS SDK as below. [Read Docs](#)

Setup



Environment

SDK SOURCE

JS

```
1 <script src="https://sdk.cashfree.com/js/v3/c
2
3 <script>
4   const cashfree = Cashfree({
5     mode: "sandbox"
6   });
7 </script>
```

Copy Code

Here we need to choose the setup option I am going with the **CDN URL**. If you are using React choose **NPM Package**

You will get this HTML code there:

```
<script src="https://sdk.cashfree.com/js/v3/cashfree.js"></script>

<script>
  const cashfree = Cashfree({
    mode: "sandbox"
  });
</script>
```

This code will go on your index.html

3. Create Order

To process any payment on Cashfree PG, the merchant needs to create an order in the cashfree system.
Use Payment Session ID from the generated response to render the checkout page. [Read Docs](#)

Environment

Sandbox

Amount

100

Phone Number

8474090589

Order Options

- ☒ Basic Order
- ☒ Return URL
- ☐ Customer Details
- ☐ Notify URL (Webhooks)
- ☐ Payment Methods
- ☒ Order Expiry
- ☐ Order Note
- ☐ Order Tags

Curl Node Go PHP Java .NET Python

```
5 Cashfree.Environment = Cashfree.Environment.SANDBOX;
6
7 var request = {
8   "order_amount": 100.00,
9   "order_currency": "INR",
10  "order_id": "devstudio_7520356",
11  "customer_details": {
12    "customer_id": "devstudio_user",
13    "customer_phone": "8474090589"
14  },
15  "order_meta": {
16    "return_url": "https://www.cashfree.com/devst
17  },
18  "order_expiry_time": "2024-12-28T17:41:18.144Z"
19 };
20
21 Cashfree.PGCreateOrder("2023-08-01", request).then((r
22   console.log('Order created successfully:', respons
23 }).catch((error) => {
24   console.error('Error:', error.response.data.messa
25 });
```

Copy Code

Try Now

Response of the API will appear
here.
Set environment, your app id and
your secret key

At this stage, you will get lots of options to choose from. You can select all of them if you want.
I'm adding Order Expiry here to show you how to work with it.

Here I want you to use your API keys. Make sure you add them in the .env file.

I have made some modifications to this code.

Create a cashfree-service.js file like this:

```
/* cashfree-service.js

const { Cashfree } = require("cashfree-pg"); // You have to install this package
// Setup
Cashfree.XClientId = process.env.CASHFREE_APP_ID;
Cashfree.XClientSecret = process.env.CASHFREE_SECRET_KEY;
Cashfree.XEnvironment = Cashfree.Environment.SANDBOX;

// This function will generate session ID for us so, we can make payment
exports.createOrder = async ( orderId, orderAmount, orderCurrency, customerID,
customerPhone ) => {
  try {
    const expiryDate = new Date(Date.now() + 60 * 60 * 1000); // 1 hour from now
    const formattedExpiryDate = expiryDate.toISOString();

    const request = {
      order_amount: orderAmount,
      order_currency: orderCurrency,
      order_id: orderId,
      customer_details: {
        customer_id: customerID,
        customer_phone: customerPhone,
      },
      order_meta: {
        // This will make a GET API call on your site
        return_url: `http://localhost:3000/payment-status/${orderId}`,
      },
      Order_expiry_time: formattedExpiryDate, // Set the valid expiry date
    };

    const response = await Cashfree.PGCreateOrder("2023-08-01", request);

    return response.data.payment_session_id;
  } catch (error) {
    console.error("Error creating order:", error.message);
  }
};
```

As you can see I added this into a function. So I can call this in my controller.js.

Note: In the “return_url” we need to give your API endpoint to check the order status.
We are going to create it in the upcoming steps.

4. Load Checkout

Enter the Payment Session ID below to open the checkout page. [Read Docs](#)

Payment Session ID

session_o4Aq-pygXVf6c6FMFPgS7A8RZb59HAtMtietNoz0AzG_7AdkslaFvUB0oLK-rku6TbIGpGqv0tnsxFyoz09foTjwy0Jh28BSmsOfNbsOuDiLUnGA67bA7Jl

Environment

Sandbox

Steps

- ☐ 1. Integrate Cashfree SDK
- ☐ 2. Initialize **cashfree** variable
- ☐ 3. Trigger Cashfree **checkout()**

HTML ReactJS

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1">
6   <title>Cashfree Checkout Integration</title>
7   <script src="https://sdk.cashfree.com/js/v3/cashfree.js"></script>
8 </head>
9 <body>
10 <div class="row">
11 <p>Click below to open the checkout page</p>
12 <button id="renderBtn">Pay Now</button>
13 </div>
14 <script>
15   const cashfree = Cashfree({
16     mode: "sandbox",
17   });
18   document.getElementById("renderBtn").addEventListener("click", () => {
19     let checkoutOptions = {
20       paymentSessionId: "session_o4Aq-pygXVf6c6FMFPgS7A8RZb59HAtMtietNoz0AzG_7AdkslaFvUB0oLK-rku6TbIGpGqv0tnsxFyoz09foTjwy0Jh28BSmsOfNbsOuDiLUnGA67bA7Jl"
21     };
22     cashfree.checkout(checkoutOptions);
23   });
24 </script>
25 </body>
26 </html>

```

Copy Code

Click below to open the checkout page in current tab

Pay Now

Here now we are going to set up our frontend code.

1. Add CDN in the head tag.
`<script src="https://sdk.cashfree.com/js/v3/cashfree.js"></script>`
2. Add a payment button in your HTML code.
3. Initialize cashfree variable like this in your script tag:

```
const cashfree = Cashfree({
  mode: "sandbox",
});
```
4. We need to add an event listener on the payment button on click.
5. On event trigger, we need to make an API call on our backend so we will get the `payment_session_id`.
6. The last work in our frontend code is to trigger `Cashfree checkout()`.

Use this code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Cashfree Checkout Integration</title>
  <script src="https://sdk.cashfree.com/js/v3/cashfree.js"></script>
</head>
<body>

  <button id="renderBtn">Pay Now</button>

  <script>
const cashfree = Cashfree({
  mode: "sandbox",
});

document.getElementById("renderBtn").addEventListener("click", async () => {
  try {
    // Fetch payment session ID from backend
    const response = await fetch("http://localhost:3000/pay", {
      method: "POST",
    });

    const data = await response.json();
    const paymentSessionId = data.paymentSessionId;

    // Initialize checkout options
    let checkoutOptions = {
      paymentSessionId: paymentSessionId,
      redirectTarget: "_self",
    };

    // Start the checkout process
    await cashfree.checkout(checkoutOptions);

  } catch (err) {
    console.error("Error:", err);
  }
});
</script>
</body>
</html>
```

Use your endpoint accordingly.

```
cashfree.checkout(checkoutOptions);
```

This will be one of the payment options.

Once it is done, the backend code will call another API.

In my case this is the endpoint:

`return_url: `http://localhost:3000/payment-status/${orderId}``,

Let's see what we need to add in the backend now.

Server Side

5. Get Payments for an Order

Once payment is attempted, Customer will be redirected to return_url provided in Create Order API (Step 3).
Use the below API to fetch status of all transactions for an order, analyse the API Response and show the final payment status to your customer. [Read Docs](#)

Environment

Sandbox

Order ID

devstudio_53464828

CurlNodeGoPHPJava.NETPython

```
4
3 Cashfree.XClientId = "TEST430329ae80e0f32e41a393d78b923034";
4 Cashfree.XClientSecret = "TESTaf195616268bd6202eeb3bf8dc458956e7";
5 Cashfree.XEnvironment = Cashfree.Environment.SANDBOX;
6
7 - Cashfree.PGOrderFetchPayments("2023-08-01", "your-order-id").then
8   console.log('Order fetched successfully:', response.data);
9 - }).catch((error) => {
10   console.error('Error:', error.response.data.message);
11   });
```

Copy CodeTry Now

Response of the API will appear here.
Set environment, your app id and your secret key

Payment States

☐ Success

Paid Transaction

☐ Pending

Late Authorization

☐ Failure

User Dropped

Transaction Failure

Node JS

```
1 let getOrderResponse = []; //Get Order API Response
2 let orderStatus;
3
4 - if(getOrderResponse.filter(transaction => transaction.payment_status
5   orderStatus = "Success"
6 - }else if(getOrderResponse.filter(transaction => transaction.payment
7   orderStatus = "Pending"
8 - }else{
9   orderStatus = "Failure"
10 }
```

Copy Code

Payment Status will appear here once the API is triggered

This is the last step where you check what happened with the order (the user's payment). It was Success or Failure. For this, we need to add this code in our “cashfree-service.js”:

```
/* cashfree-service.js

exports.getPaymentStatus = async (orderId) => {
  try {
    const response = await Cashfree.PGOrderFetchPayments("2023-08-01", orderId);

    let getOrderResponse = response.data;
    let orderStatus;

    if (
      getOrderResponse.filter(
        (transaction) => transaction.payment_status === "SUCCESS"
      ).length > 0
    ) {
      orderStatus = "Success";
    } else if (
      getOrderResponse.filter(
        (transaction) => transaction.payment_status === "PENDING"
      ).length > 0
    ) {
      orderStatus = "Pending";
    } else {
      orderStatus = "Failure";
    }

    return orderStatus;

  } catch (error) {
    console.error("Error fetching order status:", error.message);
  }
};
```

This function will take an order ID and return the order status.

Now it's up to you have you are going to use it.

- You show this to your front end by sending a response.
- You can store it in the database for records.

**That's all about Cashfree integration.
Happy Coding!**